

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: BTECH

Branch : Electronics and Computer Engineering/ Electronics and Computer Science

Semester : VII

Subject Code & Name: BTECPC703 DevOps (Development & Operations)

Max Marks: 60

Date: 10/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. Objective type questions. (Compulsory Question)			12
1	Which of the following is NOT a core principle of DevOps?		1
	a. Collaboration b. Automation c. Isolation d. Continuous improvement	L2	
2	What is the main goal of the DevOps delivery pipeline?		1
	a. Code obfuscation b. Continuous integration and delivery c. Manual testing d. Centralized deployment	L1	
3	Which of these tools is commonly used in the DevOps lifecycle?-		1
	a. Docker b. Jenkins c. Kubernetes d. All of the above	L1	
4	What does CI stand for in DevOps?		1
	a. Continuous Integration b. Cloud Infrastructure c. Collaborative Innovation d. Code Injection	L2	
5	A DevOps engineer is expected to possess which of the following skills?		1
		L3	

	a. Network configuration	b. Software development	c. Infrastructure automation	d. All of the above		
6	In GIT, which command is used to view the commit history?				L2	1
	a. git commit	b. git log	c. git status	d. git push		
7	Why is DevOps needed on the cloud?				L1	1
	a. To ensure scalability of applications	b. To simplify collaboration across teams	c. To enhance automation and monitoring	d. All of the above		
8	Which of the following is a popular cloud service provider?				L1	1
	a. Amazon Web Services (AWS)	b. Azure	c. Google Cloud Platform (GCP)	d. All of the above		
9	In the context of CAMS, "blamelessness" refers to:				L2	1
	a. Avoiding accountability	b. Encouraging organizational learning by avoiding finger-pointing	c. Reducing automation failures	d. Ignoring root cause analysis		
10	What does "CAMS" stand for in DevOps?				L1	1
	a. Culture, Automation, Measurement, Sharing	b. Configuration, Automation, Metrics, Scaling	c. Collaboration, Agile, Metrics, Security	d. Code, Analysis, Monitoring, Scaling		
11	What is Docker primarily used for?				L2	1
	a. Operating system installation	b. Containerization of applications	c. Virtual machine management	d. Network configuration		
12	Which of the following is a key component of Kubernetes?				L2	1
	a. Pods	b. Containers	c. Repositories	d. Virtual machines		

Q.2	Solve the following.		12
A)	Explain the core principles of DevOps in detail. How do these principles contribute to improving software delivery processes?	L2	6
B)	What are Continuous Integration (CI) and Continuous Delivery (CD)? Highlight their significance in the DevOps lifecycle.	L1	6
Q.3	Solve the following.		12
A)	What is cloud computing? Discuss its role in modern IT infrastructure and its relevance to DevOps practices.	L2	6
B)	How do cloud service providers like AWS, Azure, and Google Cloud enable and enhance DevOps practices?	L3	6
Q.4	Solve Any Two of the following.		12
A)	What is version control, and why is it essential in modern software development?	L2	6
B)	Explain the importance of Git in the industry. Discuss its benefits and how it supports collaborative software development.	L2	6
C)	Discuss commonly used Git commands for managing repositories, such as clone, add, commit, and push.	L1	6
Q.5	Solve Any Two of the following.		12
A)	Discuss the role of automation in DevOps. How does continuous delivery automation help deliver high value?	L2	6
B)	What is the significance of sharing in the CAMS framework? Discuss how sharing knowledge and practices improves team collaboration and learning.	L1	6
C)	What is configuration management in DevOps? Discuss its role in ensuring consistency and efficiency in software environments.	L2	6
Q.6	Solve Any Two of the following.		12

A)	Explain the history of software engineering. How has the field evolved from early programming to modern software development practices?	L1	6
B)	Describe the role of Jenkins in Continuous Integration (CI). How does Jenkins automate the software build and deployment process?	L2	6
C)	Explain how Docker, Kubernetes, Jenkins, and Ansible work together in a DevOps pipeline. Discuss their interactions and how they contribute to a streamlined deployment process.	L2	6
*** End ***			